

Age-related changes in PD-1 expression coincide with increased cytotoxic potential in Vδ2 T cells during infancy

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ARTICLE INFO

Keywords:

Gammadelta T cells
Cord blood
Infant
PD-1
BCG
Cytotoxic potential

ABSTRACT

Human Vγ9Vδ2 T cells respond to several diverse pathogens by sensing microbial cholesterol intermediates. Unlike CD4 T cells, they are poised for rapid Th1-like responses even before birth, which allows them to play a key role in the first line of defense against pathogens in early life. However, their regulation and functional maturation during infancy (in particular the acquisition of cytotoxic potential) remain understudied. We thus characterized their responses to cholesterol intermediates and *Bacille Calmette-Guérin* in a cohort of African neonates and 12-month-old infants. Infant Vδ2 lymphocytes exhibited intermediate or adult-like expression of markers associated with differentiation or function, intermediate proliferative responses, and adult-like cytotoxic potential. The enhancement of Vδ2 cell cytotoxic potential coincided with decreasing PD-1 and increasing NKG2A expression. Our results are consistent with the hypothesis that switching from a PD-1⁺ to a NKG2A⁺ phenotype during infancy indicates a shift in mechanisms regulating Vδ2 T cell function.

1. Introduction

Before birth, CD4 T cells are inherently skewed in favor of regulatory or Th2 responses over Th1 responses [1]. While this promotes the maintenance of fetal-maternal tolerance by limiting pro-inflammatory responses, the unique characteristics of the fetal immune system also contribute to high susceptibility to infections in early life (reviewed in [2,3]). In addition, due to the fetal immunologic milieu, prenatal priming by microbial antigens in some cases results in non-protective immune memory, that may blunt subsequent protective responses and further increase the risk of infections in infancy [4–10]. In contrast to CD4 T cells, the innate-like Vγ9Vδ2 (Vδ2) T cells are already poised for Th1 responses during fetal life [11]. In addition, they can rely on cytokines of myeloid origin (IL-23 and IL-15) to support proliferation, thus

mounting CD4-independent responses to pathogens [11–13]. Because of these characteristics, Vδ2 cells likely play a key role against infections early in life, but their responses and regulation in infancy are still understudied.

Vδ2 T cells, a subset of γδ T lymphocytes, mount responses against a wide range of microbial agents (including *mycobacteria* and *P. falciparum*) due to their unique reactivity. They monitor cholesterol biosynthesis by sensing the levels of its metabolic intermediates, specifically high potency molecules produced by a variety of microbes, and low-potency compounds produced by host cells [14–16]. These small molecules, which are characterized by a pyrophosphate moiety, are collectively called phosphoantigens (PAg), and include the microbial E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMBPP), as well as the host-derived Isopentenyl-pyrophosphate (IPP) [14,16,17].

Abbreviations: Vδ2, Vgamma9Vdelta2 T cells; TCR, T cell receptor; PD-1, programmed cell death 1; BCG, *Bacille Calmette-Guérin*; PAg, phosphoantigens; IPP, isopentenyl-pyrophosphate; CB, cord blood; CBMC, cord blood mononuclear cells; PBMC, peripheral blood mononuclear cells; ZOL, zoledronic acid monohydrate; IL-2, interleukin 2; SI, stimulation index.

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<https://doi.org/10.1016/j.cellimm.2020.104244>

Received 29 June 2020; Received in revised form 8 October 2020; Accepted 30 October 2020

Available online 13 November 2020

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Aminobisphosphonate drugs are also stimulatory [18], but act indirectly, by blocking cholesterol biosynthesis and inducing intracellular accumulation of IPP [19].

V δ 2 cell responses are not restricted by known MHC molecule, [20,21] thus they are most likely independent of immunogenetic factors. PAg-mediated activation requires a V γ 9JP rearrangement paired with the V δ 2 chain [22,23]. Gamma chains containing the V9 gene but not the JP segment display low or no reactivity to PAg [22]. The molecule BTN3A1 is critical for sensing of PAg via an inside-out mechanism that has been partly clarified, but the structural details of ligand recognition by the V γ 9V δ 2 TCR are still under investigation [24–29].

V δ 2 T cells are present at low frequencies in cord blood and expand in the periphery during the first few years of life in response to microbial exposure [30]. While *in vitro* stimulation elicits qualitatively similar responses in neonatal and adult V δ 2 cells, the magnitude of the responses is generally lower for cord blood cells. Our group and others demonstrated that V δ 2 cells in cord blood display lower proliferation in response to PAg than their adult counterparts, but expand robustly following aminobisphosphonate stimulation [11,12,31]. In addition, the V δ 2 cell phenotype in cord blood is typical of a mostly naïve population compared to their adult counterpart [11,12,32,33], and possibly suggests an alternate regulatory mechanism for these cells in early life [34]. We have shown that neonatal V δ 2 cells express the inhibitory receptor PD-1 for longer periods than their adult counterparts after *in vitro* activation, and this molecule may serve as a key regulatory factor in early life.

V δ 2 peripheral expansion during childhood is likely accompanied by the acquisition of adult-like broad antimicrobial function. Investigations aimed at monitoring this process by probing V δ 2 T cell responses in the first months of life with common stimuli are scarce, even after recent studies on $\gamma\delta$ T cells in pediatric malaria and cytomegalovirus infection [35,36]. To expand our understanding, we compared V δ 2 T cell responses to aminobisphosphonates and live *Bacille Calmette-Guérin* (BCG) at birth and at 12 months of age (longitudinal specimens) to responses in healthy adults. We included live BCG as a model microorganism because this vaccine is routinely administered at birth to neonates in Sub-Saharan Africa and produces the microbial PAg, thus triggering V δ 2 T cells, but limited information is available about V δ 2 T cell contribution to BCG responses in neonates and infants [12,37,38]. We observed that V δ 2 T cells at 12 months of age have intermediate differentiation and functional characteristics, but are overall more reminiscent of the mature, adult cells than of their cord blood counterparts.

2. Materials and methods

2.1. Specimen collection and mononuclear cell isolation

Cord blood (CB) specimens were obtained from uncomplicated full term pregnancies at the Ndirande Government Health Center, in Blantyre, Malawi. CB samples were collected from HIV-negative women who had been enrolled in a clinical trial (ClinicalTrials.gov Identifier: NCT01443130). The specimens included in the current study were collected from deliveries of HIV uninfected women who also had no evidence of malaria infection during pregnancy. Some of the infants born to women in the clinical trial were followed up for one year after birth [5]. Infant peripheral blood (2–3 ml) was collected at 12 months of age in EDTA sterile tubes. The clinical trial and study procedures were approved by the Institutional Review Board of the University of Maryland School of Medicine and by the College of Medicine Research Ethics Committee of the University of Malawi College of Medicine in Blantyre, Malawi.

Adult peripheral blood specimens were purchased from the New York blood bank or collected from healthy volunteers at the University of Maryland School of Medicine after obtaining a signed informed consent form from each participant in the context of a protocol approved by the Institutional Review Board.

Cord blood was collected from the umbilical vein of delivered placentas soon after uncomplicated births. Umbilical cord blood (15–60 ml) was collected using a sterile cord blood collection unit (Pall medical, Port Washington, NY) after wiping the cord to remove maternal blood and sterilizing the collection site with ethanol. Cord blood or peripheral blood was diluted with phosphate buffered saline, PBS (Lonza, Walkersville, MD) and layered over Lymphocyte Separation Medium (LSM, Corning, NY) density gradient to isolate cord blood or peripheral blood mononuclear cells (CBMC and PBMC respectively). CBMC or PBMC were frozen in 90% fetal bovine serum (FBS, Gibco, Thermo Fisher Scientific, Waltham, MA), 10% DMSO (Millipore Sigma, St. Louis, MO) freezing medium. All specimens were stored at -130°C or below before use.

A total of 18 neonate, 25 infant and 18 adult donor specimens were utilized for this study. Few specimens were not included in all experiments due to limited cell availability.

2.2. Cell culture

After thawing, CBMC or PBMC pellets were incubated for two minutes at room temperature (RT) with 50 μg of DNase (Stem Cell Technologies, Vancouver, Canada), then resuspended in RPMI 1640 supplemented with 10% FBS (Gibco, Thermo Fisher Scientific), 2 mM L-glutamine, and 10 $\mu\text{g}/\text{ml}$ gentamicin (Gibco, Thermo Fisher Scientific) for counting. Cell number was determined by staining with anti-CD45 for six minutes at room temperature and counting the events in the mononuclear cell gate on a Guava (Millipore Sigma). After counting, cell concentration was adjusted to 1.5×10^6 cells/ml. In order to expand V δ 2 T lymphocytes, cultures were treated with zoledronic acid monohydrate (ZOL, Millipore Sigma) at 0.5 μM , or BCG at the pre-optimized multiplicity of infection (MOI) of one (Pasteur strain, provided by Dr. Manganeli), in the presence of 100 IU/ml of human recombinant interleukin 2 (IL-2) (Tecin, NIH reagent program, NIH, Bethesda, MD). IL-2 alone (100 IU/ml) was used as negative control treatment. After thawing, BCG was resuspended in complete medium and centrifuged at 4000g for 15 min. 3×10^6 colony forming units (CFU) were added to each CBMC well, and $(0.9) \times 10^6$ CFU to each PBMC well. Cells were incubated for 14 days at 37°C with 5% CO_2 and fresh cytokine was added every 3 days; one volume of fresh medium was added on days 7 and 10. On days 14 and 16, a fraction of the CBMC was used to determine V δ 2 T cell frequency, phenotype and function.

2.3. Flow cytometry

Ex vivo or cultured CBMC were resuspended in PBS, stained with a fixable viability dye (BD Horizon™ 780, BD bioscience, Franklin Lakes, NJ or Zombie Aqua, Biolegend, San Diego, CA) at room temperature for 15 min, and washed twice with PBS-10% FBS. Afterwards, cells were stained at 4°C for 15 min with the directly conjugated monoclonal antibodies listed below. After washing with PBS, cells were resuspended in PBS-0.3% paraformaldehyde. $(3-8) \times 10^5$ lymphocytes (gated on the basis of forward and side scatter profiles) were collected for each specimen on an AriaII or LSRII (BD Biosciences). The acquisition files were analyzed with FlowJo software (FlowJo LLC, Ashland, OR).

The monoclonal antibodies used for polychromatic cell surface staining are listed below. Anti-V δ 2 (clone B6), anti-PD-1 (clone EH12.2H7), anti-CD39 (clone A1), anti-CD69 (clone FN50), anti-CD3 (clone OKT3, UCHT1), anti-CD16 (clone 3G8), anti-CD27 (clone O323), anti-CD56 (5.1H11), were purchased from Biolegend (San Diego, CA). Anti-CD70 (clone Ki-24) was purchased from BD Biosciences. Anti-CD3 (clone BW264/56), anti-PD-1 (clone PD1.3.1.3), anti-CD16 (clone REA423), anti-CD45RA (clone T6D11), anti-CD45RO (clone UCHL1), anti-CD56 (clone REA196), anti-NKG2A (clone REA110), anti-NKG2D (clone BAT221), anti-CD62L (clone 145/15), anti-CD28 (clone 15E8) were obtained from Miltenyi Biotec (Bergishch Gladbach, Germany). Anti-CD25 (clone CD25-4E3), and anti-HLA-DR (clone L243) were purchased from Thermo Fisher Scientific.

2.4. Production of cytotoxic mediators

In order to evaluate V δ 2 T cell cytotoxic potential, we monitored the production of cytotoxic mediators in response to stimulation. Sixteen days after stimulation, the cells were counted and aliquoted at $(4-6) \times 10^5$ per tube. Following viability and surface staining (as described above), the cells were permeabilized by incubation with fixation/permeabilization solution (BD Biosciences) at 4 °C for 20 min. After two washes with 1X Perm/wash buffer (BD Biosciences), the cells were incubated for 40 min at room temperature with anti-Perforin (clone DG9, Thermo Fisher Scientific), anti-Granulysin (clone DH2, Biolegend), and anti-Granzyme A (clone CB9, Biolegend) or anti-Granzyme B (clone GB11, Biolegend) diluted at preoptimized concentrations in 1X Perm/wash buffer (BD Biosciences). Finally, the cells were washed once with Perm/wash buffer and resuspended in buffer for acquisition. At least 10^5 lymphocytes were collected for each sample on a FACS Aria II (BD Biosciences).

2.5. Granule mobilization assay

Sixteen days after stimulation, CBMC or PBMC were resuspended at 2×10^6 cells/ml in fresh complete medium and re-stimulated in 96-well plates pre-coated with anti- $\gamma\delta$ TCR (clone B1.1, Thermo Fisher Scientific). The plates were coated overnight at 4 °C with anti- $\gamma\delta$ TCR (diluted 1:100 in PBS, 50 μ L/well). Cells were plated in triplicate (100 μ L/well) in presence of anti-CD107a AlexaFluor488 (3 μ L/well, clone H4A3, Biolegend), GolgiPlug (brefeldin A, 1 μ g/ml, BD Biosciences), and GolgiStop (monensin, 1 μ g/mL, BD Biosciences). After a 6-hour incubation, the cells were collected, washed once with cold PBS, and stained with a viability dye, followed by surface staining. At least 10^5 lymphocytes were collected for each sample on a FACS LSR II (BD Biosciences).

2.6. RNA extraction, RT-PCR

RNA from $(1-3) \times 10^6$ cells *ex vivo* and 16 days after stimulation was extracted by the Direct-zol RNA miniprep (Zymo Research, Irvine, CA) as directed by the manufacturer. Total RNA (1 μ g) was converted into cDNA using the iScript cDNA Synthesis kit (Bio-Rad Laboratories, Hercules, CA), according to manufacturer's instruction. cDNA was diluted with one volume of nuclease-free water. Polymerase chain reaction of V γ 9 fragment was performed as described [39], using the following primers for the V γ 9 chain (V γ 2 according to an alternate nomenclature): oligo-V γ 9 (5'-ATC AAC GCT GGC AGT CC-3') and oligo-C γ -1 (5'-GTT GCT CTT CTT TTC TTG CC-3'). The PCR products were purified using Genomic DNA clean & Concentrator-10 (Zymo Research) for the run-off reaction.

2.7. Run-Off reaction

Primer extension reactions were performed as described [39]. Each reaction contained (3–6) μ L of PCR product, 3 mM MgCl₂, 0.2 mM dNTP, 0.1 mM 6-carboxyfluorescein (6-FAM)-labeled primer (C γ -6: 5'-6-FAM-AAT AGT GGG CTT GGG GGA AAC-3'), 0.15 units GoTaq Hot Start polymerase (Promega, Madison, WI), and buffer master mix. Each sample was made up of run-off products (3 μ L), formamide (7 μ L) (Applied Biosystems, Foster City, CA) and 1 μ L GeneScan 500 LIZ size standard (Thermo Fisher Scientific). After a denaturation step (5 min at 95 °C followed by immediate quenching on ice), products were loaded on a 3130 genetic analyzer (Applied Biosystems) and run on a performance-optimized polymer (POP-7). Molecular size and relative frequency of extension products were determined using GENEMAPPER software (Applied Biosystems). To standardize the data irrespective of the run-off primer position, CDR3 length variation was expressed in terms of the total V γ 9 coding region lengths. Run-off product lengths were corrected by adding the length of the known mRNA coding region outside the run-off primer-binding site.

2.8. Statistical analysis

Statistical analyses were performed using the software GraphPad Prism V5.0f (GraphPad Software, La Jolla, CA). For each variable, D'Agostino & Pearson omnibus normality test was employed to assess whether values were normally distributed. Differences between means (for normally distributed variables) or medians (for variables displaying a non-Gaussian distribution) were evaluated, respectively, using a student's *t*-test or Mann-Whitney test for unpaired groups, and paired *t*-test or Wilcoxon test for paired groups. Comparisons between multiple groups we performed using ANOVA or Kruskal-Wallis test. Age groups were always considered unpaired (as in addition to neonate-infant pairs we included a few infants with no paired neonatal specimens), while phenotypic subsets of V δ 2 T cells were treated as paired groups.

3. Results

3.1. Infant V δ 2 T cells proliferate more than neonatal cells, but less than adult cells in response to ZOL and live BCG

Neonatal V δ 2 T cells mount weaker responses to phosphoantigens than adult cells [11,12,31]. To test how their responses change during infancy, we compared V δ 2 T cell differentiation and functional characteristics for CBMC, 12-month-old infant PBMC (mostly paired specimens, with few additional unpaired samples) and reference adult PBMC in responses to stimulation with zoledronic acid (ZOL) plus IL-2 or live BCG plus IL-2. IL-2 alone represented the negative control treatment.

Ex vivo, infant V δ 2 T cell frequency (reported as % of CD3⁺ cells) was significantly higher than cord blood frequency, but still lower than adult levels (Fig. 1A). Two weeks after stimulation, V δ 2 T cells significantly expanded with both treatments (compared to *ex vivo* and IL-2 treatment), but their frequency was significantly lower after BCG compared to ZOL stimulation, for all age groups (Fig. 1B, 1C). However, the difference between ZOL and BCG was more noticeable for neonates than for infants, and marginal for adults. The average frequency of V δ 2 T cells after ZOL treatment was approximately 5, 2 and 1.3 fold the frequency they reached after BCG stimulation for neonates, infants and adults respectively (12.3% versus 2.5% for neonates, 44.7% versus 23.8% for infants, 82.0% versus 65.4% for adults). To normalize the results for the low-level proliferation driven by IL-2, we calculated the stimulation index (SI, # of V δ 2 T cells after ZOL (or BCG) / # of V δ 2 T cells after IL-2). This parameter confirmed that the difference in proliferative responses to ZOL and BCG is most marked for neonates (Fig. 1D). In fact, only neonates had a significantly lower SI in response to BCG compared to ZOL; for infants the difference did not reach statistical significance and for adults the two SI were comparable. The SI confirmed that infant V δ 2 T cells proliferate more than their cord blood counterparts in response to both ZOL and BCG, and suggested that IL-2-driven proliferation contributes to the differences in V δ 2 T cells frequencies between infants and adult. In fact, after normalizing proliferative responses for IL-2 effects, the ZOL and BCG SI for infants and adults were comparable (Fig. 1D).

For some CBMC specimens the frequency of V δ 2 T cells was similar after BCG or IL-2 stimulation. We thus decided to assess whether BCG was expanding PAg-sensing V δ 2 cells or whether the IL-2 added to BCG-treated cultures was driving non-specific proliferation of other V δ 2 cell subsets. We thus performed run-off analysis (also called spectratyping) of CBMC to determine the length distribution of the CDR3 region in V γ 9 chains after culture and the approximate proportion of PAg-sensing V γ 9 chains. The V δ 2 cells that can sense PAg express a V9-JP rearrangement in the gamma chain paired with the V δ 2 chain [22]. Most of these rearrangements generate chains in the 990–996 nucleotide range [40], and, likewise, most V γ 9 chains in the 990–996 nucleotide range carry a V9-JP rearrangement [39]. Thus, the proportion of chains in the 990–996 nucleotide range (% 900–996) approximates the proportion of PAg-sensing chains in the V γ 9 pool. Run-off analysis of CBMC revealed

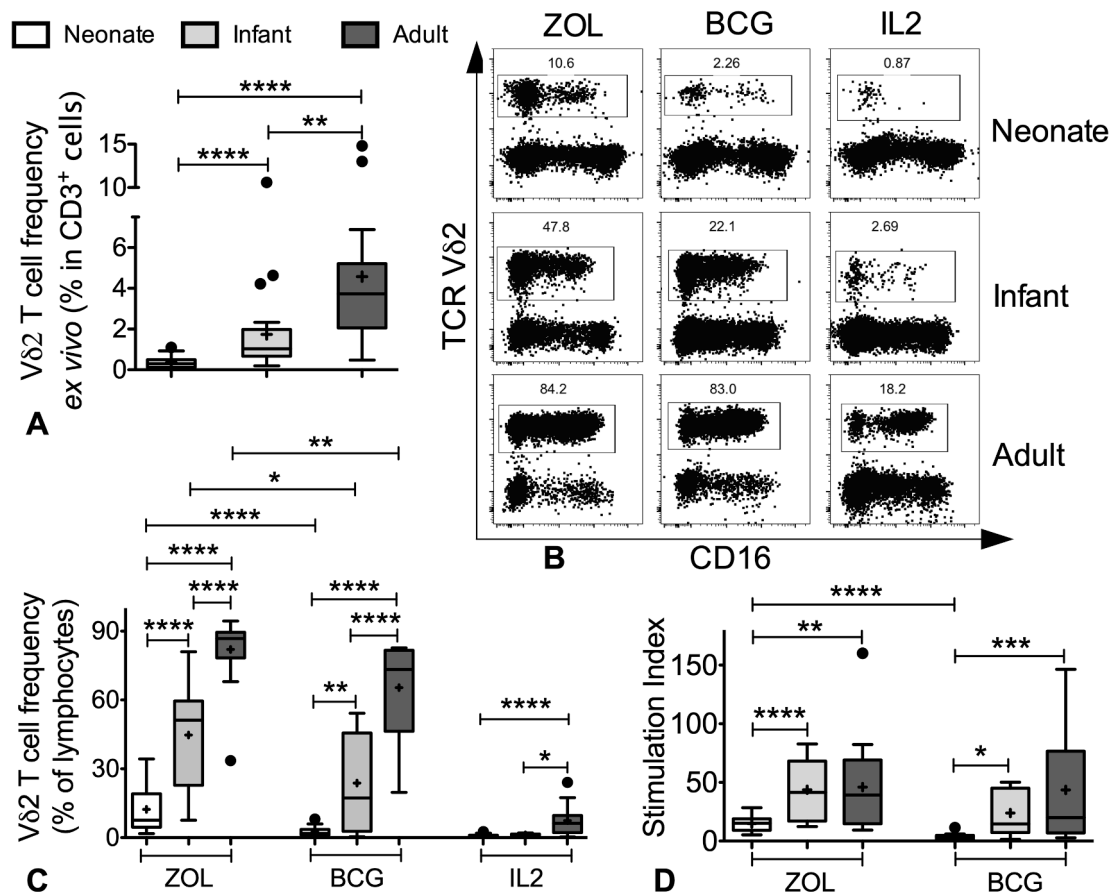


Fig. 1. Proliferative responses to Zoledronate or live BCG change with age. Vδ2 T cell frequency was determined for neonatal (cord blood), infant and adult specimens *ex vivo* and after expansion. CBMC and PBMC were treated with IL-2 (100 U/ml), ZOL (0.5 μM) + IL-2, or BCG (MOI = 1) + IL-2 for 14 days, then flow cytometry analysis was performed. (A) The box plot shows mean, median and IQR for *ex vivo* Vδ2 T cell frequency across the three age groups. (B) The dot plots depict the frequency of Vδ2 T cells after expansion for a representative individual from each age group (with paired neonate and infant specimens). (C) The box plot shows mean, median and IQR for Vδ2 T cell frequency across the three age groups for all treatments. (D) The stimulation indices (SI) were calculated as the number of Vδ2 T cells after ZOL or BCG stimulation divided by the number of Vδ2 T cells after IL-2 treatment. The box plots show mean, median and IQR for SI across the three age groups in response to ZOL and BCG. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

that the % 990–996 was highest following ZOL treatment (consistent with higher SI and Vδ2 T cell frequencies) but BCG stimulation yielded a significantly higher proportion of PAg-sensing clones than IL-2 (Supplementary Fig. 1). Conversely, IL-2 stimulation, as we described before [39], selected clones with shorter Vγ9 chains that are not able to sense PAg levels. Infant PBMC yielded comparable results, though the % 990–996 following BCG stimulation tended to be higher in PBMC compared to CBMC cultures (data not shown).

3.2. Infant Vδ2 T cells express adult levels of PD-1 and intermediate NKG2A after ZOL stimulation

We determined the distribution of Vδ2 T cells in naïve and memory subsets, which has functional relevance since different memory compartments in adults display distinct preferential effector responses [32,41,42]. *Ex vivo*, in neonate and infants the majority of cells had a naïve phenotype (CD45RA⁺CD27⁺), with some central memory cells (CD45RA⁺CD27⁺) and few effector memory cells (CD45RA⁺CD27⁺, data not shown and Supplementary Table 1). After ZOL stimulation, the majority of neonatal Vδ2 T cells had differentiated into central memory (66.5%) with a smaller proportion of effector memory (21.2%) and virtually no terminally differentiated cells (CD45RA⁺CD27⁺CD16⁺, 2.1%, Supplementary Fig. 2 and data not shown). On average 11.1% of Vδ2 lymphocytes retained a naïve phenotype. Conversely, infant and adult Vδ2 cells had differentiated predominantly into effector memory

cells (68.4% and 84.2%, respectively), with a smaller subset of central memory cells (20.3% and 15.3%, respectively), <10% of terminally differentiated effectors (data not shown) and virtually no naïve cells (<2% data not shown). BCG stimulation resulted in a differentiation pattern comparable to ZOL, for all age groups (data not shown).

We have previously shown that neonatal Vδ2 T cells express the immune checkpoint PD-1 for a prolonged period after ZOL stimulation [34], thus we assessed whether the same is true for cells obtained from 12-month-old infants. *Ex vivo*, cord blood Vδ2 T cells had the highest proportion of PD-1⁺ cells, followed by infant and adult Vδ2 T cells ($p = 0.0002$, Supplementary Table 1). In culture, as expected, large proportions of cord blood Vδ2 T cells expressed this immune checkpoint two weeks after ZOL stimulation, while infant specimens displayed lower frequencies of PD-1⁺ Vδ2 lymphocytes, similar to their adult counterparts (Fig. 2A, 2B). For all age groups, cells stimulated with BCG displayed significantly lower frequencies of PD-1⁺ Vδ2 lymphocytes than those treated with ZOL, but the difference between treatments was most marked for neonatal specimens (Fig. 2A).

In contrast to PD-1, the expression of the inhibitory NK receptor NKG2A increased with age. *Ex vivo*, cord blood Vδ2 lymphocytes displayed barely any detectable expression of NKG2A ($5.4\% \pm 3.2\%$), as previously reported [12,43], while infant and adults had on average 41.83% and 70.2% NKG2A⁺ Vδ2 cells respectively ($p < 0.0001$, Supplementary Table 1). Similarly, after expansion the frequencies of NKG2A⁺ Vδ2 T cells were higher in infant than in neonatal specimens,

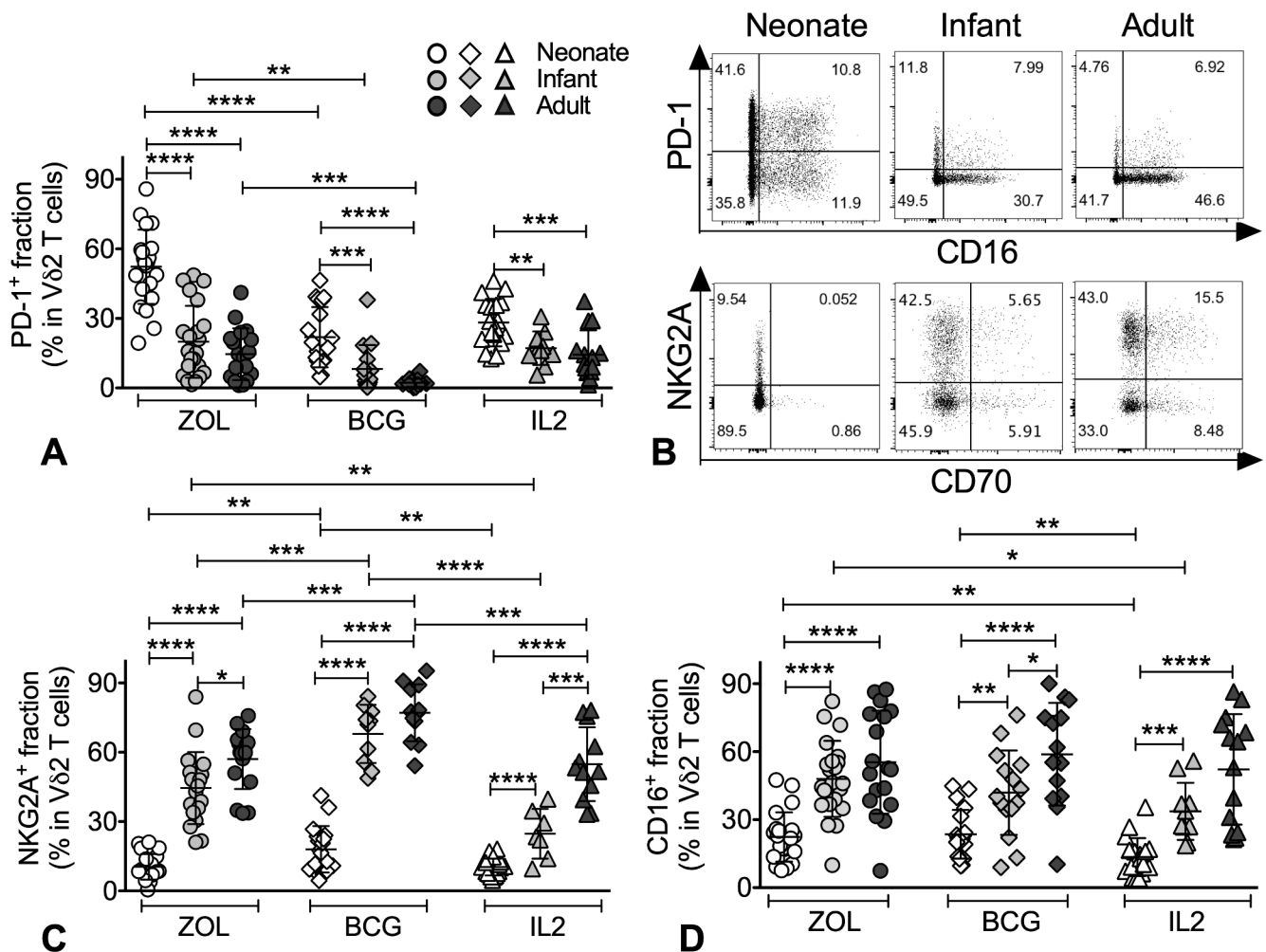


Fig. 2. Age-related differences in Vδ2 T cell expression of inhibitory receptors. Two weeks after stimulation with ZOL, BCG or IL-2, Vδ2 T cell expression of membrane markers was assessed by flow cytometry. (A) The scatter plot depicts the frequency of PD-1⁺ Vδ2 T cells for individual donors, as well as mean and SD for each group of specimens. (B) The dot plots show PD-1, NKG2A and CD16 expression following ZOL stimulation for a representative donor for each age group (with paired neonate and infant specimens). (C-D) The scatter plots depict, respectively, the frequency of NKG2A⁺ and CD16⁺ Vδ2 T cells for individual donors, as well as mean and SD for each group of specimens. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

and highest in adult PBMC (Fig. 2B and 2C). Interestingly, BCG treatment induced a larger subset of NKG2A⁺ cells compared to ZOL for all age groups (Fig. 2C).

Infant PBMC also displayed elevated frequencies of CD16⁺ Vδ2 T cells compared to CBMC after all treatments (Fig. 2B and 2D). In particular, infant Vδ2 lymphocytes expressed CD16 at adult levels after ZOL (and IL-2) but not BCG stimulation. Increased CD16 expression is likely to have functional consequences, as this activating Fcγ receptor mediates antibody-dependent cell cytotoxicity (ADCC).

3.3. Infant Vδ2 T cells have higher cytotoxic potential than neonatal cells

NKG2A expression is thought to be possibly associated with acquisition of cytotoxic capacity [43]. We thus compared the frequencies of Vδ2 T cells producing cytotoxic mediators *ex vivo* and after expansion across all age groups. *Ex vivo*, most infant and adult Vδ2 T cells expressed perforin (on average 81.8% and 79.4% of total Vδ2 T cells, respectively), while the frequency of perforin⁺ Vδ2 T cells in cord blood was low (on average < 10%), consistent with previous reports (Fig. 3A, left panel) [44–46]. Conversely, the frequency of granzyme⁺ cells among circulating Vδ2 T cells in infants was significantly lower than in adults, but significantly higher than in neonates (Fig. 3A, right panel).

After stimulation, the proportion of perforin⁺ Vδ2 T cells was higher

in infant than in neonate specimens for all treatments, reaching adult levels after ZOL and IL-2 stimulation but not after BCG treatment (Fig. 3B and 3C). Granzyme⁺ Vδ2 T cells were also significantly more frequent in infant than neonatal specimens after ZOL stimulation, while they were present at comparable frequencies in the two age groups following BCG or IL-2 treatment (Fig. 3D). Granzyme⁺ Vδ2 T cells in infants reached frequencies comparable to adults only in response to ZOL (Fig. 3D). Finally, ZOL and BCG stimulation yielded comparable frequencies of perforin⁺ Vδ2 cells for all age groups (Fig. 3C). The majority of Vδ2 T cells (more than 80%) expressed both granzyme A and B in all age groups (data not shown).

The improved ability of infant Vδ2 T cell to produce cytotoxic mediators coincided with decreased PD-1 and increased NKG2A expression (Figs. 2 and 3). Consistent with this, after ZOL stimulation, perforin⁺ cord blood Vδ2 T lymphocytes were enriched in the PD-1⁺ subset compared to the PD-1⁻ subset and total Vδ2 cell population. For infants, there was a small but significant difference between the PD-1⁻ and the PD-1⁺ subset, while no difference between subsets was found for adults (Fig. 4A and 4B). Perforin⁺ cells were also enriched in the CD16⁺ subset compared to CD16⁻ and total Vδ2 cells, in both neonatal and infant specimens, but the difference between subsets was more marked for neonatal specimens (Fig. 4C and 4D). BCG stimulation yielded similar results (Supplementary Fig. 3A and 3B for PD-1 and CD16 subsets,

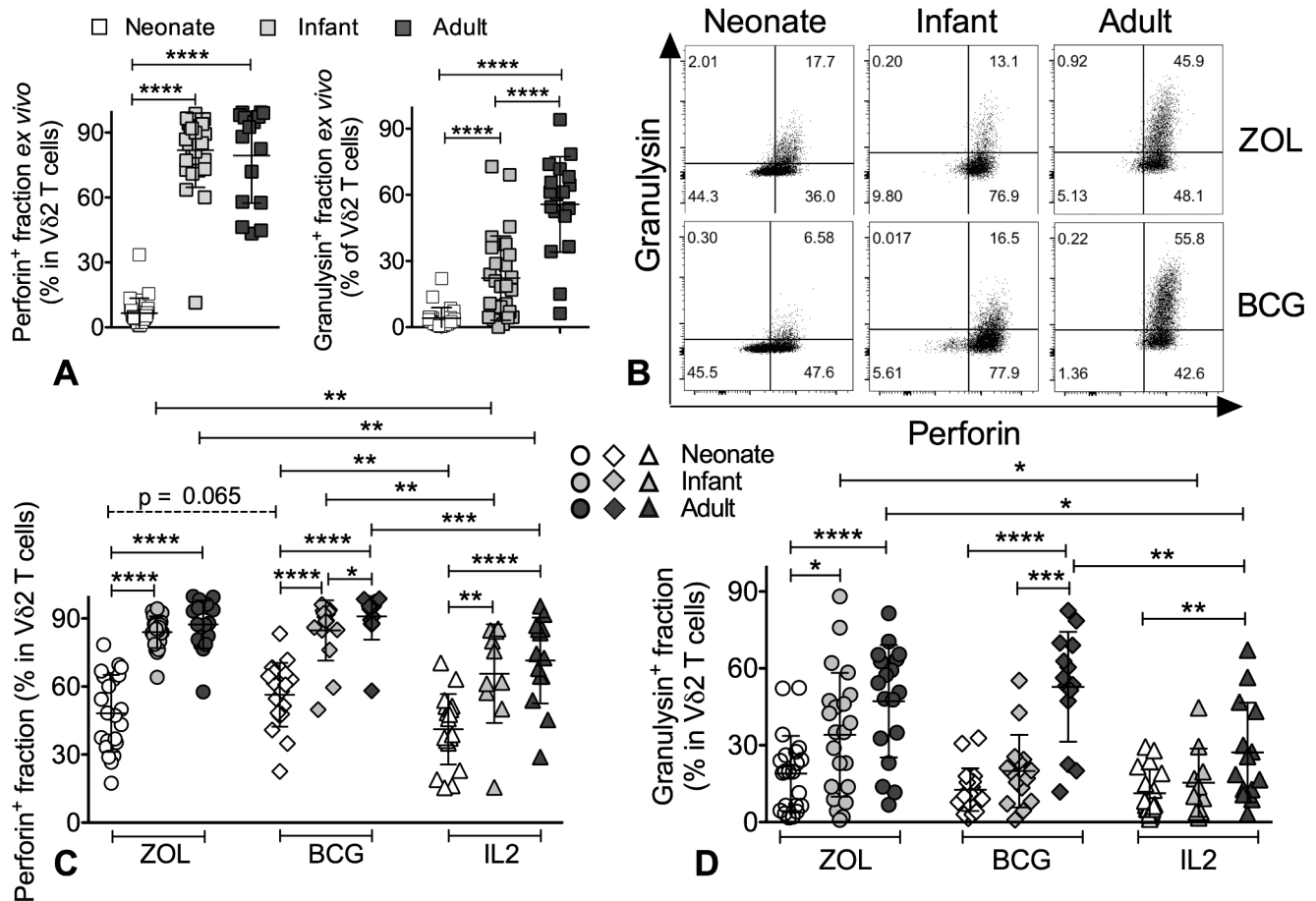


Fig. 3. Vδ2 T cell production of cytotoxic mediators increases with age. The proportion of Vδ2 T cells storing cytotoxic mediators was assessed *ex vivo* and 16 days after stimulation by intracellular staining. (A) The scatter plots show individual frequencies, mean and SD of Vδ2 T cells containing perforin and granulysin *ex vivo* for the three age groups. (B) The dot plots show the frequency of perforin⁺ and granulysin⁺ Vδ2 T cells after stimulation with ZOL or BCG for a representative individual from each age group (with paired neonate and infant specimens). (C, D) The scatter plots depict, respectively, the frequency of perforin⁺ and granulysin⁺ Vδ2 T cells for individual donors, with mean and SD for each group of specimens. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

respectively). Granulysin production in Vδ2 cells did not differ based on PD-1 expression (data not shown). Conversely, after both ZOL and BCG stimulation the CD16⁺ subset contained a significantly higher frequency of granulysin⁺ Vδ2 cells than the CD16⁻ subset in all age groups (Supplementary Fig. 3C and data not shown).

In order to evaluate Vδ2 T cells cytotoxic potential, 16 days after the initial treatment we restimulated the cell cultures for 6 h with plastic immobilized anti-γδ TCR and monitored cytotoxic granule mobilization by CD107a staining. A larger proportion of ZOL- and BCG-treated infant Vδ2 T cells mobilized cytotoxic granules (CD107a⁺) compared to their cord blood counterpart, reaching levels of degranulation comparable to adult cells (Fig. 5A and 5B). Cord blood CD107a⁺ Vδ2 T lymphocytes were enriched in the PD-1⁻ subset compared to the PD-1⁺ subset and the total Vδ2 cells population, while in infants and adults specimens there were no differences between subsets (Fig. 5C and 5D). The frequency of CD107a⁺ Vδ2 T cells in cord blood specimens tended to be higher after BCG than after ZOL stimulation, but the difference did not reach statistical significance. Conversely, for adult and infant specimens, the frequency of CD107a⁺ Vδ2 T cells was lower after BCG than after ZOL, with a statistically significant difference for adults but not for infants (Fig. 5A).

4. Discussion

This study shows that circulating Vδ2 T cells in 12-month-old infants display adult-like cytotoxic potential and stronger proliferative

responses than cord blood cells. This functional differentiation is accompanied by phenotypic changes that are consistent with higher cytotoxic capacity. Our results are consistent with existing literature on γδ T cells in neonates and young children, and help elucidate the timing and features of peripheral Vδ2 T cell maturation in early life.

We and others have previously demonstrated that cord blood Vδ2 T cells have some unique characteristics compared their adult peripheral blood counterparts, including differentiation state and lower responsiveness to PAg [11,12,33]. Vδ2 T cells appear to undergo postnatal peripheral expansion [30,47], driven by cholesterol intermediates produced by intestinal microbiota or other environmental microbes. This process, which takes place over the first few years after birth [30], is thought to result in a gradual acquisition of adult-like features. Learning more about Vδ2 T cell in infancy is key, as their ability to mount rapid Th1-like responses [46] and proliferate independently of CD4 T cell help [11,13,39] makes them an essential component of the first line of defense against pathogens in early life. For example, Jagannathan and colleagues demonstrated that Vδ2 T cell responses to malaria correlate with protection from infection in Ugandan children with repeated exposures to *P. falciparum* [35]. However, few studies investigated functional maturation of γδ T cells in healthy infants [36], particularly in reference to proliferative responses and cytotoxic potential, prompting us to compare these features for neonatal (cord blood) and infant (12-month-old peripheral blood) Vδ2 T cells. While cord blood, in general, does not seem to mirror neonatal peripheral blood [48], a recent paper demonstrated that γδ T cells in cord blood are comparable to γδ T

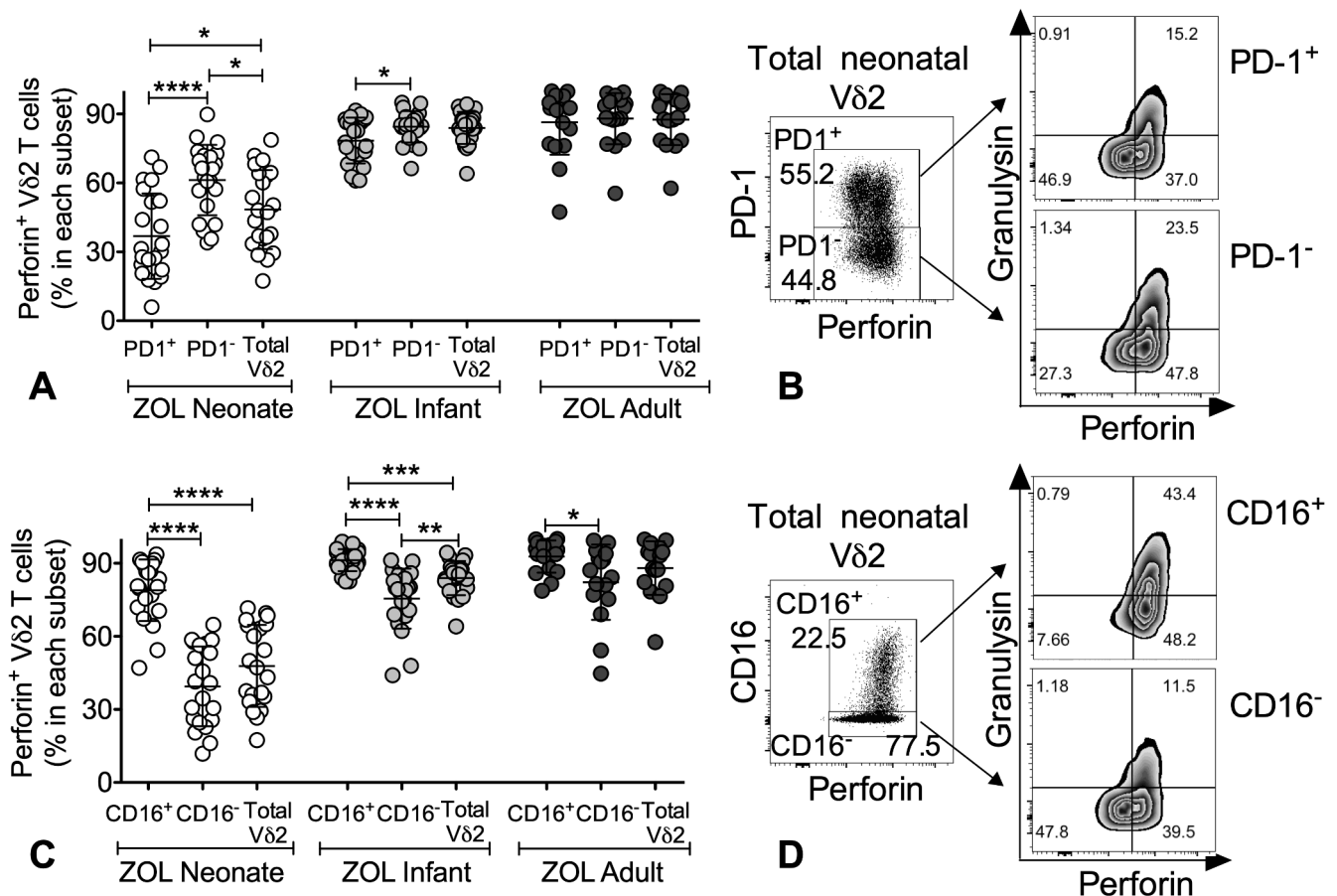


Fig. 4. Cells storing cytotoxic mediators are enriched among PD-1⁻ and CD16⁺ Vδ2 T lymphocytes. Sixteen days after stimulation the production of cytotoxic mediators was assessed by intracellular staining. (A) The scatter plot depicts the frequency of perforin⁺ Vδ2 T cells for individual donors, as well as mean and SD for each group of specimens, comparing the PD-1⁺ with the PD-1⁻ subset and the total Vδ2 population. (B) The zebra plots show the frequency of perforin⁺ and granulysin⁺ cells in the PD-1⁺ and PD-1⁻ Vδ2 T lymphocyte subset for a representative neonatal specimen after ZOL stimulation. (C) The scatter plot depicts the frequency of perforin⁺ Vδ2 T cells for individual donors, with mean and SD for each group of specimens, comparing the CD16⁺ subsets with the CD16⁻ subset and the total Vδ2 population. (D) The zebra plots show the frequency of perforin⁺ and granulysin⁺ cells in the CD16⁺ and CD16⁻ Vδ2 T cell subset for a representative neonatal specimen after ZOL stimulation. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

neonatal peripheral blood [36] and can thus be used as a reference to study the maturation of Vδ2 T cells.

Compared to cord blood Vδ2 T cells, infant peripheral blood Vδ2 T cells displayed stronger proliferation in response to both ZOL and BCG (though not at adult levels), elevated cytotoxic potential, and changes in NKG2A, CD16, and PD-1 expression. Specifically, increased frequencies of Vδ2 T cells with strong cytotoxic potential (perforin⁺ or CD107a⁺) in infant specimens coincided with reduced PD-1⁺ expression. The observation that increased cytotoxic potential at one year of age coincides with reduced PD-1 levels supports our previous hypothesis that prolonged PD-1 expression by cord blood Vδ2 T lymphocytes may be associated with control of their function, in particular cytotoxicity, during fetal and early postnatal life [34]. Also consistent with this hypothesis, the proportions of perforin⁺ and CD107a⁺ cells were higher in the PD-1⁻ subset than in the PD-1⁺ subset of Vδ2 T lymphocytes. Differences between PD-1⁺ and PD-1⁻ subsets were very marked for neonatal Vδ2 T cells, less marked but significant for infant cells, and not noticeable for adult cells. Increased cytotoxic potential in infants also coincided with higher frequencies of NKG2A⁺ Vδ2 cells, in agreement with a recent report showing that high CD94/NKG2A expression identifies the cytotoxic subset in adults [49].

Interestingly, BCG stimulation yielded significantly lower frequencies of PD-1⁺ Vδ2 T cells and total Vδ2 T cells than ZOL treatment. There are multiple potential explanations for this difference. PD-1 may behave as an activation marker and persists on activated cells [50]. If

this is the case, BCG may result in a weaker stimulation compared to ZOL and induce lower activation levels that have mostly subsided by the time we assess cell phenotype, with consequent downmodulation of PD-1. Alternatively, since BCG stimulation induces 5–10 fold the amount of IFN-γ present in ZOL-stimulated culture (data not shown), higher IFN-γ concentrations may result in significantly elevated PD-L1 expression levels [51,52] in BCG-stimulated cultures compared to ZOL-treated cultures. PD-1 engagement by its ligand PD-L1 may drive a negative selection of the PD-1⁺ cells (possibly slowing down their cell cycle, inducing apoptosis or both) and thus bias the proliferating Vδ2 lymphocytes towards a PD-1⁻ population. Prolonged PD-1 expression in neonatal, but not infant and adult Vδ2 cells may thus help explain differences in Vδ2 population size across age groups following BCG stimulation. Finally, differences in the cytokine milieu not related to IFN-γ (for example, TGF-β concentrations [53]) in BCG versus ZOL-stimulated cultures may be responsible for variations in PD-1⁺ Vδ2 T cell frequencies. Future studies employing PD-L1 blocking and IFN-γ neutralizing antibodies will help test these hypotheses.

It is important to note that BCG stimulation of CBMC, while resulting in a relatively weak Vδ2 T cell expansion compared to ZOL, drove differentiation of cells with high cytotoxic potential. Surprisingly, this population of cytotoxic Vδ2 effectors maintains a central memory phenotype at least *in vitro*, and thus would be expected to retain the ability to proliferate upon subsequent early life stimulation [41,42]. Conversely, in 12-month-old infants, most of the perforin⁺ and CD107a⁺

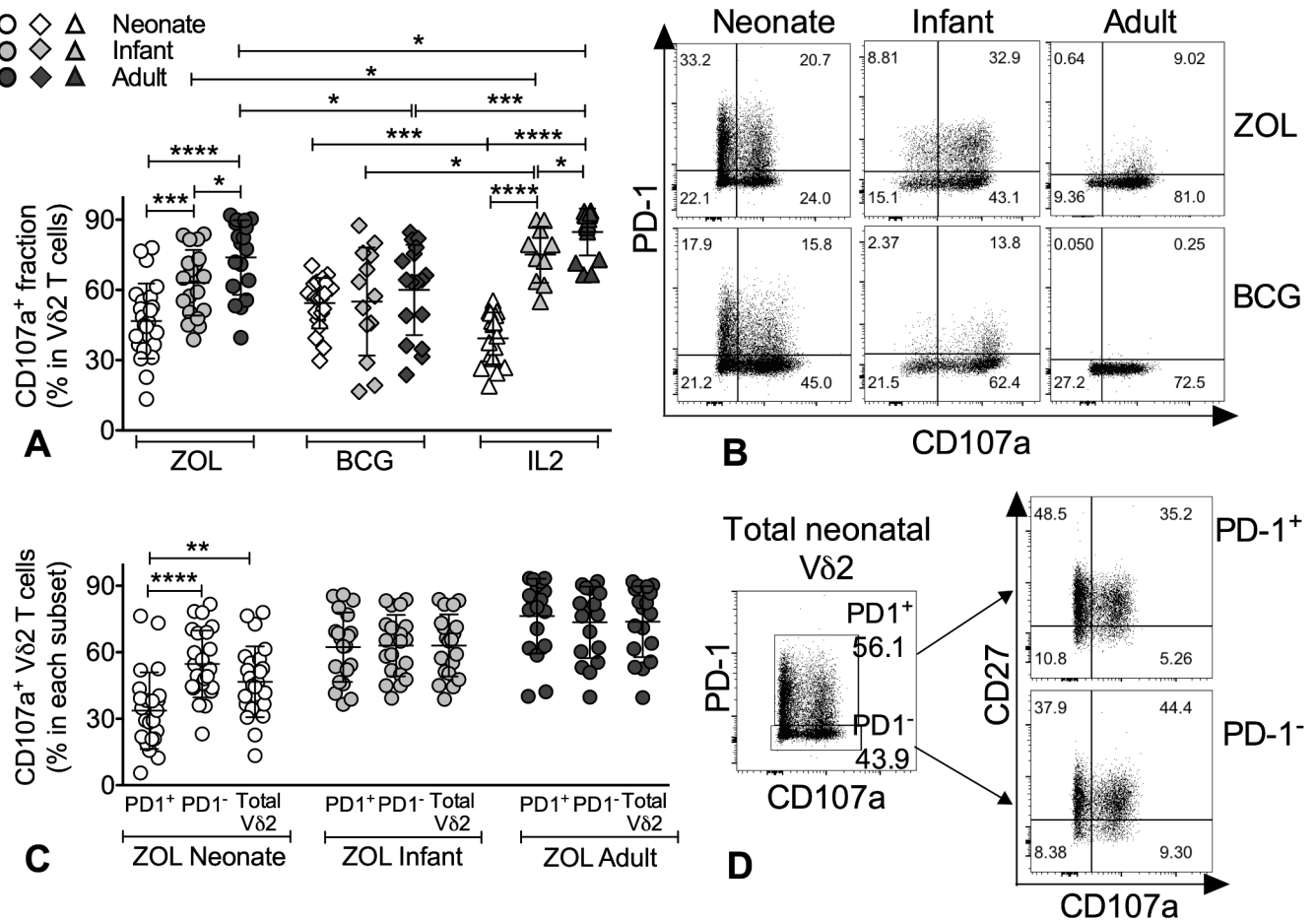


Fig. 5. Degranulating cells are enriched among PD-1⁻ and CD16⁺ Vδ2 T lymphocytes. Sixteen days after stimulation, CBMC or PBMC were restimulated for 6 h with plastic-immobilized anti γδ TCR (clone B1.1) in the presence of GolgiPlug (1 μg/ml), and GolgiStop (1 μg/mL). (A) The scatter plot depicts the individual frequencies of CD107a⁺ Vδ2 T cell, with mean and SD for each group of specimens. (B) The dot plots show the frequency of CD107a⁺ Vδ2 T cells for a representative donor from each age group after ZOL and BCG stimulation (with paired neonate and infant specimens). (C) The scatter plot shows the individual frequencies of CD107a⁺ Vδ2 T cells with mean and SD for PD-1⁺, PD-1⁻ and total Vδ2 T cells. (D) The dot plots display the frequency of CD107a⁺ cells in the PD-1⁺ and PD-1⁻ Vδ2 T cell subset for a representative neonatal specimen after ZOL stimulation. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

cells have acquired an effector memory phenotype, associated with lower proliferative potential [41,42].

While the general conclusions of our study are likely applicable to infants across all countries, differences between cohorts living in distinct geographic locations cannot be excluded. Immunogenetic factors are unlikely to influence Vδ2 T cell responses and maturation, since no highly polymorphic molecules are directly involved in PAg sensing. However, infants in low resource settings are likely to experience more diverse exposure to microbial agents and more frequent infections than infants in developed countries, which may bias or accelerate Vδ2 T cell maturation in early life. *P. falciparum* is known to elicit Vδ2 T cell responses, thus we monitored malaria acquisition in the infant cohort during follow up [5]. Since only one of the infants included in this study tested positive to *P. falciparum* before 12 months of age, we can assume that malaria was not a major confounder in our investigation. We were, however, unable to screen for other common pediatric infections, thus we cannot exclude that other pathogens contributed to modify Vδ2 T cell maturation. Another potential limitation of this study is that the healthy adults serving as controls were a racially diverse pool of individuals living in North America. This does not represent an ideal control group, because microbial exposure in North America and Africa is substantially different. However, enrolling healthy Malawian adults would have been difficult in the absence of routine screening for parasitic, viral and bacterial infections that are often subclinical in adults in developing countries.

Overall, Vδ2 T cells in 12-month-old infants are functionally more similar to adult than neonatal cells, though some of their features are intermediate. A process of postnatal maturation driven by microbial PAg is most likely responsible for the differences observed between cord blood and infant peripheral blood Vδ2 T cells. However, there may be other contributing factors. A recent study highlighted that γδT cells in the fetal and the postnatal period derive from two distinct thymocyte populations, identifiable by specific repertoire features [47]. This observation is consistent with previous evidence that T cells at birth and in adulthood derive from different hematopoietic stem cell populations [54]. Based on these reports, Vδ2 T lymphocytes at birth and in infancy may be a mixture of cells derived from a fetal and a postnatal thymocyte population, each with distinct functional features. A more robust representation of fetal-derived cells at birth than at 12 months of age would thus contribute to the observed age-related differences in Vδ2 T cell function. In future studies we will seek to determine the relative contribution of PAg-driven maturation versus thymocyte precursor origin to Vδ2 T cell regulation and function in infants, to help devise strategies that harness the antimicrobial potential of these cells to improve immune responses against pathogens and, possibly, vaccines in early life.

Funding

This work was supported by National Institute of Health (grant

R01AI104702).

Author contribution

HH executed the experiments, analyzed the data and wrote the manuscript; SB was instrumental for CB collection and discussion of the clinical data; GM was responsible for processing and cryopreservation of CBMC and PBMC; DR contributed to data analysis; FB, GD, and RM cultured BCG, prepared the stocks and determined the CFU; THD and RGM managed the daily infant follow up operations in Malawi; MKL supervised the infant follow up and provided critical insights for manuscript revision; CC designed the experiments, contributed to data analysis and wrote the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the Ndirande Health Center, Blantyre, Malawi for assistance with cord blood and infant peripheral blood collection. We thank Dr. Karl B. Seydel and the Molecular Core at the University of Malawi, College of Medicine, in Blantyre, Malawi for the outstanding technical and logistic support. We are very grateful to the IHV and UMGCCC Flow Cytometry Core for their support with Vδ2 T cell analyses.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cellimm.2020.104244>.

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